



NORBERT MUZILA

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ABOUT ME

Emerging Space Systems Engineer who recently completed a Master's degree in Space Studies (**Space Engineering and Applications**) at International Space University, with **hands-on experience** in the design, integration, prototyping, and testing of CubeSats, robotic systems, UAVs, and space mission architectures. Experienced in systems engineering practices including subsystem integration, AIT workflows, concurrent engineering design, verification and validation, and environmental testing methodologies. Technical exposure spans satellite bus and payload systems, launch operations, embedded technologies, and ground segment software development. Adept at working in multidisciplinary and international engineering environments, with a strong interest in applying **AI and digital technologies to next-generation aerospace** and complex **engineering systems**. Seeking a **Space Engineering** or data opportunity systems or software engineering, aerospace, robotics, and advanced technology development.

KEY SPACE ENGINEERING PROJECTS

CubeSat 1U - Project-Based Learning (PBL) | PROJECT LEAD

- Coordinated a team of **8 students** on 1U CubeSat, subsystem integration scheduling, interface management, and integration review and was the lead the Software subsystem.
- Led** full hardware integration of a 1U CubeSat following CubeSat Design Specification (CDS) standards: EPS (electrical power system), OBDH (on-board data handling), UHF/VHF communications module, ADCS sensors, and structural assembly.

Robotics Engineering Project - Autonomous System Design and Integration | CHIEF PROGRAMMER

- Designed and integrated a robotic system with embedded sensor suites, motor control electronics, and communication interfaces; developed Python-based control software for autonomous navigation.
- Applied hardware-software integration techniques: real-time sensor data acquisition, actuator control, and system-level
- functional verification on bench-level test setups.

Team Project - Planetary Rover and Drone System Design | AI/SOFTWARE LEAD

- Led** the **AI/SW** subsystem in a multi-disciplinary team project(20 students) designing a Mars duo system for sub-surface exploration mission.
- Performed structural and electronics integration on the **rover prototype**; conducted functional tests to verify actuator, sensor, and communication subsystem performance.

EDUCATION AND TRAINING

01/09/2025 – 24/07/2026 Illkirch-Graffenstaden, France

MASTERS OF SPACE STUDIES (MSS) SPACE ENGINEERING AND APPLICATIONS TRACK International Space University

01/09/2020 – 27/07/2025 Harare, Zimbabwe

BACHELOR OF SCIENCE (BSC.) HONOURS DEGREE IN DATA SCIENCE AND SYSTEMS University of Zimbabwe

Specialization - Machine Learning, Deep Neural Networks, Big Data Analytics, Statistical Analysis, Predictive Modelling, and Software Systems Design, Software Development

SKILLS

Model Based System Engineering (MBSE) | Cameo SysML | STK | Git/Github, Docker, Gitlab | SOLID WORKS | Elastic Search / Kibana | Python | SQL | Docker | Microsoft Excel | Power BI | R | Fusion 360 | C++ | MATLAB

● **WORK EXPERIENCE**

DATA SCIENCE RESIDENCE RESEARCHER – ZIMBABWE CENTRE FOR HIGH PERFORMANCE COMPUTING – 01/12/2024
– 31/07/2025 – HARARE , ZIMBABWE

- Conducted applied research in **Artificial Intelligence and Big Data Analytics**, with a focus on optimizing machine learning algorithms for efficient performance on large-scale datasets.
- Performed **performance evaluation** on large datasets, including operational data from **Zabbix** and supported data-driven decision to C-Level.
- Leveraged **High-Performance Computing (HPC)** and cloud computing environments to accelerate computation intensive data processing, analytics, and machine learning model execution.
- Awarded a **Certificate of Competence in AI and Big Data Analytics** following intensive technical training in advanced data analysis and ML

STRATEGY AND DIGITAL TRANSFORMATION INTERN – STEWARD BANK – 01/08/2022 – 31/08/2023 – HARARE , ZIMBABWE

- Developed **AI/ML** algorithms for customer churn prediction, **improving account retention by 45%**.
- Automated CI/CD pipelines using **Docker and GitLab**, reducing deployment time by **90%** across test and production.
- Designed and developed interactive **Python(Jupyter Notebook)** and Power BI dashboards for channel performance analysis and executive reporting.
- Improved development efficiency by **70%** by analyzing logs via Elastic search and visualizing real-time system errors in Kibana.

● **PROFESSIONAL CERTIFICATES**

Executive Space Course – International Space University (October, 2025)

AI and Big Data - Zimbabwe Centre for High Performance Computing(August, 2025)

AWS Cloud Technical Essentials (Coursera)

Google Advanced Data Analytics Capstone (Coursera)

Introduction to Cybersecurity (Cisco Networking Academy)

● **LANGUAGE SKILLS**

Other language(s): **ENGLISH: FULL PROFESSIONAL PROFICIENCY | SHONA: NATIVE | FRENCH: BEGINER**

● **VOLUNTEERING**

CURRENT
SGAC SG[France] 2026 - Sponsorship Organizing Team Member

CURRENT
NASA OSDR AI/ML & Hardware WGs
